Alexander Hawkes

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AI/ML Engineer | GenAI, LLM Systems, and Scalable ML Pipelines

AI/ML Engineer with experience building production ready machine learning systems and LLM-powered applications. Skilled in developing scalable data pipelines, deploying models via APIs, and implementing retrieval-augmented generation (RAG) systems. Strong background in Python, cloud-based ML (AWS), and end-to-end ML lifecycle — from data processing to model deployment.

SKILLS

Languages: Python (Expert), SQL, C++, JavaScript, C#

Machine Learning: PyTorch, TensorFlow, Scikit-Learn, Model Evaluation

GenAI / LLMs: LangChain, RAG, OpenAI/Anthropic APIs, VectorDBs, Pinecone/Chroma,

FIASS

MLOps & Cloud: AWS (SageMaker, Bedrock), Docker, MLFlow, Databricks, Snowflake, CI/CD

Backend / Deployment: FastAPI, REST APIs

WORK EXPERIENCE

Alternative Stretch 06/2024 - 10/2025 Systems Engineer - **Contractor London**

- Automated operational workflows using Python, reducing manual processing time by 40% • Built scalable scripts for data handling and system monitoring across distributed environments
- Developed internal tools and APIs to support data-driven decision making

Paladin Commercial Ltd 02/2019 - 06/2021 Data Engineer - **Full-time London**

- Automated internal reporting workflows using Python, reducing processing time by 20% within 3 months, enhancing team productivity.
- Enhanced onboarding efficiency by 25% within 3 months by automating workflows, streamlining processes, and reducing manual tasks.

EDUCATION

MSc Computer Science with Artificial Intelligence
University of York 03/2025 - 03/2027

BSc (Hons) Digital Design
Brunel University London 09/2021 - 07/2024

PROJECTS

Customer Churn Prediction ML Pipeline & API

- Built end-to-end ML pipeline using Scikit-Learn for churn prediction on structured data
- Achieved 96% accuracy / X ROC-AUC
- Deployed model via FastAPI in Docker container
- Implemented inference pipeline for real-time predictions

Retrieval-Augmented Generation (RAG) Document Assistant

- Designed and deployed a RAG system using FAISS and LLM APIs for document-based

Q&A

- Improved answer grounding and reduced hallucinations through retrieval augmentation
- Processed and indexed large PDF datasets for semantic search
- Built end-to-end pipeline including ingestion, embedding, retrieval, and generation

CERTIFICATIONS

AWS Certified Machine Learning Engineer
Machine Learning Specialization
IBM Machine Learning Professional